D. Y. Patil College of Engineering and Technology, Kolhapur
Microsoft Learn Student Ambassador (MLSA) Club
Hands-on Session — Project Submission

Total Marks: 10 | Submission includes 2 Projects

PROJECT 1

1. Student Information

Name	Snehal Tanaji Khot
Roll Number	45
Branch & Year	SY CSE
Project Title	Calculator Pro Advanced Password Generator
Project Number (1-30)	7-5

2. Project Description

Description:
The Calculator Pro is a web-based application developed using HTML, CSS, and JavaScript that performs basic arithmetic and scientific calculations. It supports features like continuous calculations, history tracking, and error handling with an interactive user interface.

Problem Statement:
The objective of this project is to design and develop an interactive quiz application that enables users to test their knowledge in a fun and engaging way. The system should support multiple difficulty levels, provide real-time feedback, track scores, and maintain a leaderboard to enhance user experience and competitiveness.

3. Features Implemented

Core Features:
1. Performs basic arithmetic operations (+, −, ×, ÷)
2. Accepts user input through button clicks
3. Displays real-time input and calculated results

4. Clear/reset functionality to restart calculations

Advanced Features (Minimum 2):

1. Calculation history tracking and display
2. Continuous calculations without resetting input

Creative Feature:

1. Modern UI Design: A visually appealing calculator with smooth animations, gradients, and interactive button effects.
2. Scrollable History Panel: Displays previous calculations in an organized and user-friendly way.

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4. Use of GitHub Copilot

1. Comments written to guide Copilot

I wrote clear comments to describe the required features, such as:

1. “Create a calculator with basic arithmetic operations (+, −, ×, ÷)”
2. “Add scientific functions like square root and power”
3. “Implement calculation history tracking”
4. “Handle errors like division by zero”
5. “Design a responsive and modern UI”

These comments helped Copilot understand the structure and functionality of the project.

2. How Copilot helped generate code

GitHub Copilot assisted by:

1. Suggesting HTML structure for calculator layout
2. Generating JavaScript logic for handling button clicks and calculations
3. Providing functions for scientific operations like Math.sqrt() and Math.pow()
4. Helping implement history tracking and display updates
5. Speeding up CSS styling suggestions

5. Output / Result

The program runs an interactive calculator that allows users to perform basic and scientific calculations through a user-friendly interface. It displays results instantly, maintains a history of calculations, and supports continuous operations. The system also handles errors effectively, ensuring accurate and smooth performance.

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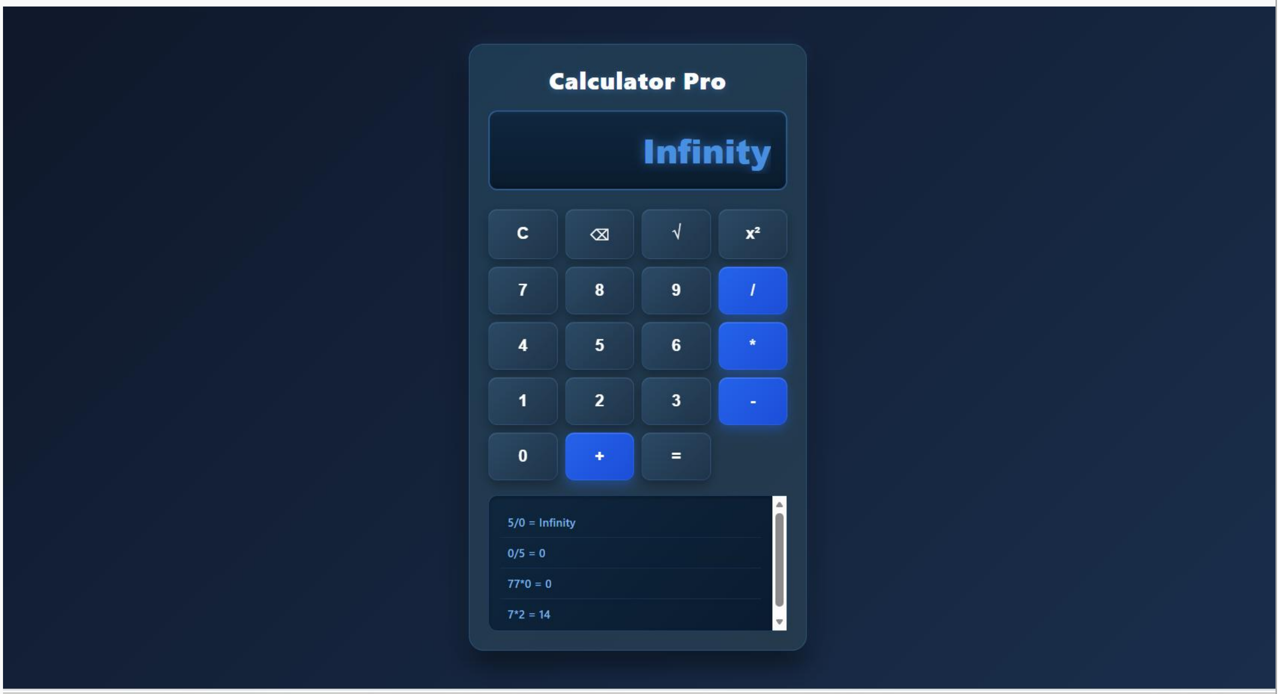
6. Screenshots / Output (Optional)



7. Project Links

GitHub Repository Link	https://github.com/Snehalkhot1/MLSA_Project
Demo / Drive Link (if any)	

Total Marks Awarded (Out of 10)	**/ 10**

PROJECT 2

1. Project Description

Description:
This project is a secure web-based application created for producing strong and unpredictable passwords. It uses a modern “cyber-style” terminal interface where users can adjust password settings such as length, symbols, and complexity, while also checking password strength instantly.

Problem Statement:

Many people use simple and common passwords that can be easily guessed or broken using brute-force attacks. Weak passwords increase the risk of unauthorized access to personal accounts and sensitive data.
This project helps solve this issue by generating highly random and complex passwords that improve account security and reduce the chances of hacking.

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2. Features Implemented

Core Features:

1. Secure password generation
2. Random strong password creation with high entropy
3. Instant password strength checking

Advanced Features (Minimum 2):

Password strength analysis
Custom password settings (length, uppercase, lowercase, numbers, symbols)
Multiple password generation at once
Export and save generated passwords to a file

Creative Feature:

1. Automatic History Storage
Generated passwords are automatically stored using LocalStorage, so users can access them even after refreshing or reopening the page.
2. **Live Security Strength Indicator**
A colorful strength meter changes dynamically from red to bright green based on password quality, giving users quick visual feedback about password safety.

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3. Use of GitHub Copilot

1. Comments Added

I added comments throughout the code to improve readability and make the program easier to understand for beginners. Comments such as `// Generate random password based on` `selected options`, `// Check password strength level`, `// Display generated password in output area`, and `// Save generated passwords to file` were included to explain the purpose of each important function. These comments help in understanding the logic step by step, make future modifications easier, and improve overall code maintenance.

2. How Copilot Assisted

GitHub Copilot helped in creating advanced CSS for the modern Neumorphic UI design, generating the password randomization logic, and writing the initial code structure for LocalStorage-based history saving.

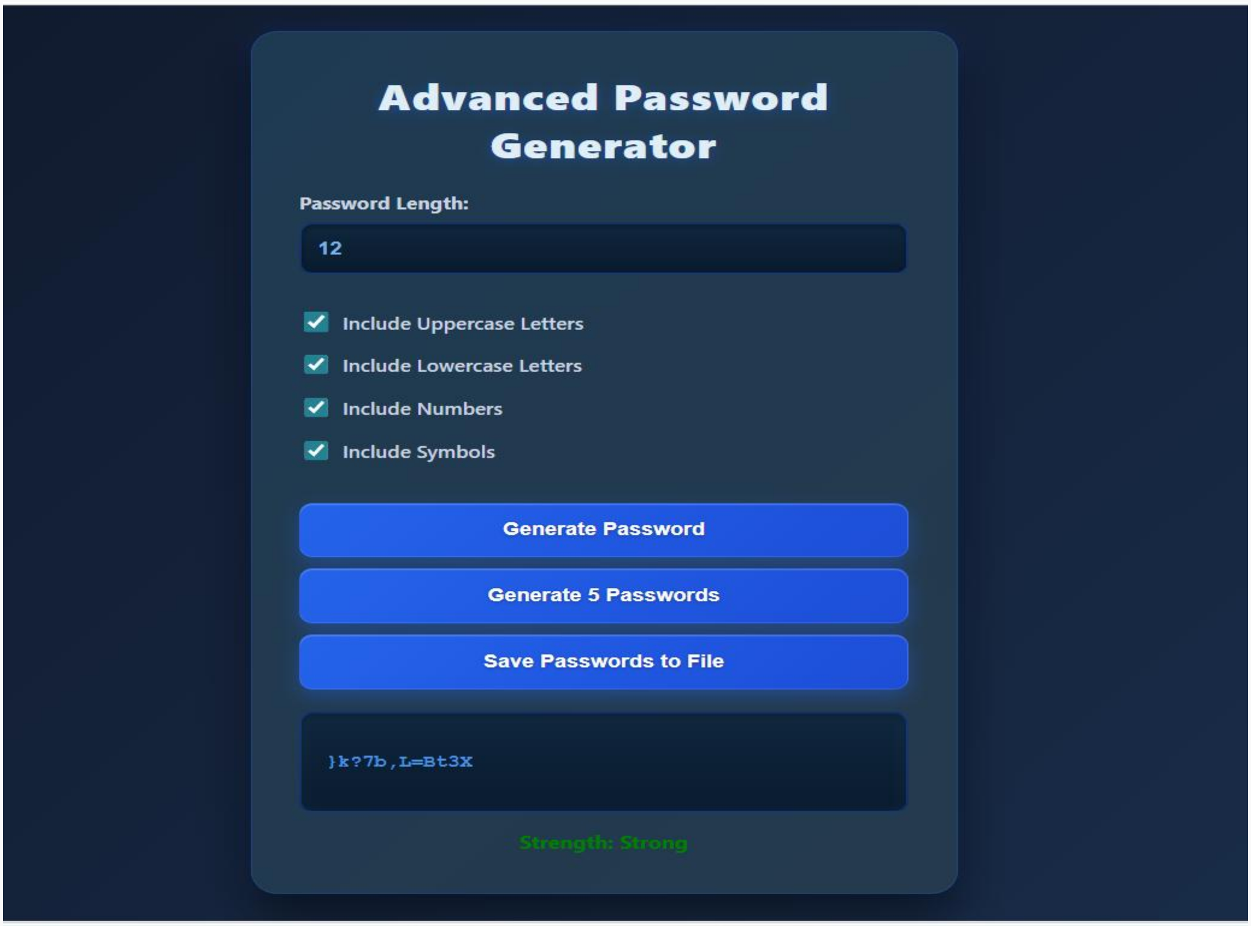
3. Changes Made Manually

I further improved the generated code by adjusting the sorting logic for better and more stable UI performance, restricting the history size to avoid unnecessary storage usage, and adding proper error handling for sharing features on desktop devices where Web Share API may not work.

4. Output / Result

The final application is a fast and visually attractive dark-themed web project that allows users to securely generate strong passwords and manage them efficiently. It gives instant feedback using strength indicators and smart suggestions, while maintaining saved history in real time. This creates a secure, responsive, and user-friendly experience for modern users.

5. Screenshots / Output (Optional)



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